INSAF ISMATH

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PROFILE

AI/ML Engineer and Researcher with experience spanning applied research and production deployment, translating business problems into end-to-end production-grade AI systems that deliver measurable impact, and publishing research at top-tier AI conferences including EMNLP and CVPR. Skilled across the full generative AI stack alongside classical ML and scalable data-led solutions.

PROFESSIONAL EXPERIENCE

9YardsX – AI and emerging technology division of 9Yards — Abu Dhabi, UAE
AI Engineer — *Sep 2026 – Present*

- **AgentOS (AI agent platform):** Leading the design and build of a self-hosted platform on which organisations build, run and govern AI agents in Arabic and English on their own infrastructure, with local models by default. Defined an agent spec that compiles to LangGraph, with a policy engine checking every step, every tool call routed through one gateway, and human approval before external actions. Built in workspace isolation and role-based access, with guardrails and security designed in from the start. *Python, FastAPI, LangGraph, PostgreSQL (pgvector), Procrastinate, vLLM, Keycloak, gVisor, React, Kubernetes, OWASP ASVS/AISVS*

2PointZero – Energy and Consumer investment platform under IHC — Abu Dhabi, UAE
AI Engineer — *Oct 2025 – Apr 2026*

- **Geospatial intelligence platform:** Drives consolidation and expansion decisions across a multi-brand salon network (130+ branches, 8 brands) via a branch-level Risk Index and hex-level Opportunity Index over an H3 grid. Shipped as an interactive dashboard and an MCP server, letting executives query the portfolio in natural language from Claude and ChatGPT. *React, MapLibre, Cesium, FastAPI, FastMCP, PostgreSQL/PostGIS*
- **Live Avatar AI Agent (Group Growth Day):** Delivered the opening and closing remarks and engaged C-suite of 20+ portfolio companies live. Grounded on per-company briefs and the Group's portfolio context, it posed tailored follow-up questions after each presentation and fielded audience Q&A within scoped guardrails. *STT-LLM-TTS pipeline, LiveKit, WebRTC, Anam 3D avatar, ElevenLabs*
- **FTE (Full-Time Equivalent) agent:** Compresses weeks of peer benchmarking into a single conversational query. ReAct Agent loop with multi-provider LLM fallback returning citation-anchored answers over a hybrid retriever on PostgreSQL, exposed as a FastMCP server. Built a per-segment iterative LightGBM imputation framework with multi-LLM consensus validation to enrich sparse peer disclosures. *FastAPI, PostgreSQL (pgvector + tsvector + pg_trgm), FastMCP, LiteLLM, Qwen3 (via Ollama), Redis*
- **Culture agent:** Multi-tenant, intent-routed LangGraph agent reinforcing company principles through structured check-in rituals, with org-scoped RAG over company docs, two-tier memory (Postgres-checkpointed state + Qdrant semantic recall), and token streaming via Celery + Redis Pub/Sub. PII-redacted Langfuse tracing with a feedback-to-score loop. In production at holding-company level, phase one of a group-wide rollout. *FastAPI, LangGraph, Qdrant, PostgreSQL, Redis + Celery, Langfuse, OpenAI Realtime API, React, WebSocket, Coolify, JWT, RBAC*
- **Agent evaluation pipeline:** Generates robustness test datasets for a multi-agent system across a 33-category question taxonomy, bootstraps ground truth by fanning each question out to three frontier models in parallel and adjudicating with an LLM-as-Judge, and exports curated datasets to Langfuse to enable continuous regression scoring of production agents. *LangGraph, Langfuse, Pydantic, asyncio*
- **Group-wide custom GPT rollout (8,000+ GPTs):** Led portfolio rollout on the Group's enterprise agentic AI platform, compressing months of manual config into an automated pipeline that LLM-maps employees to GPTs by role, function, and hierarchy and provisions at scale via Playwright. Ran technical rollout sessions with portfolio-company teams. *Prompt Engineering, Playwright*
- **Strategic research partnership (2PointZero × MBZUAI):** Represented Innovation Labs in scoping a multi-year partnership, pitched business use cases to faculty across multilingual financial NLP, multimodal document intelligence, causal decision intelligence, and AI evaluation & governance, unlocking joint publications, sponsored PhDs, and co-developed IP.

Intelligent Visual Analytics Lab (IVAL) – MBZUAI — Abu Dhabi, UAE
AI Research Engineer — *May 2025 – Oct 2025*

- **Calibration of prompt-tuned CLIP (CVPR 2026):** Contributed to a dual-regularization framework addressing underconfidence on base classes and overconfidence on novel classes; ran experiments under the standard base-to-novel protocol across 11 datasets and assisted with manuscript writing and revision. *PyTorch, CLIP, prompt-tuning frameworks (CoOp, MaPLe, KgCoOp)*

VisionLabs – Human and Object Recognition Technologies Dubai, UAE
AI Engineer *May 2024 – Aug 2024*

- **Synthetic-data face augmentation pipeline:** Trained IR-50 + ArcFace baseline on MS1M-RetinaFace-T1 (5M faces, 90K identities) and extracted IR-101 CurricularFace descriptors to compute per-identity class centers; generated per-identity synthetic faces with Arc2Face conditioned on those centers, and retraining on mixed real+synthetic data improved face-recognition accuracy by 30% across 7 benchmarks. *PyTorch; IR-50/IR-101, ArcFace/CurricularFace; Arc2Face (diffusion)*

Intelligent Visual Analytics Lab (IVAL) – MBZUAI Abu Dhabi, UAE
Graduate Research Assistant *Aug 2023 – May 2025*

- **Prompt-sensitivity benchmark for LMMs:** Built an evaluation harness on 20× NVIDIA A100s, orchestrating 1,500+ inference runs across 8 open-source and 2 proprietary LMMs over 3 MCQA benchmarks in under 2 months. Published at EMNLP 2025 Findings. *PyTorch, Hugging Face*
- **Long-form video understanding benchmark:** Built a human-in-the-loop QA generation pipeline sourcing egocentric (Ego4D) and non-egocentric videos (3–30 min) where existing benchmarks fell short; implemented inference and evaluation code across 7 open-source and 2 proprietary Video-LMMs. *PyTorch, Hugging Face*
- **Selective Channel Recalibration Attention (SCRA):** Added to ConvNeXt-Large-V2; improved fine-grained image classification accuracy on CUB Birds, FGVC Aircraft, and FoodX while staying within 5% of baseline FLOPs. *PyTorch, ConvNeXt V2*

Faculty of Engineering – University of Peradeniya Sri Lanka
Research Assistant *Jan 2021 – Aug 2023*

- **Real-time Multiple Dyadic Interaction Detection:** Built a real-time pipeline for detecting co-occurring two-person interactions in crowded surveillance footage, combining YOLOv7 + SORT for spatiotemporal localization, YOLO-Pose for skeleton extraction, and X3D-M + attention for action classification. Published at IEEE ICIIS 2023. *PyTorch; YOLOv7, SORT; X3D-M*
- **Solar GHI Nowcasting from Sky Images:** Built a physics-informed two-stage system where a classical CV sky-condition classifier routes inputs to regime-specialized ResNet sub-models, achieving 10.80% nRMSE and 98.88% Pearson correlation. Published at IEEE ICIIS 2021. *PyTorch, OpenCV; ResNet*

EDUCATION

Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI) Abu Dhabi, UAE
Master of Science in Computer Vision *Aug 2023 – May 2025*

- GPA: 3.95/4.0 (Rank 2/36); Full Scholarship (4.6% acceptance rate).
- Relevant Coursework: Mathematical Foundations of AI, Deep Learning, Probabilistic & Statistical Inference, Visual Object Recognition and Detection, Advanced Vision and Language, Human and Computer Vision

University of Peradeniya Sri Lanka
Bachelor of Science in Electrical & Electronic Engineering *Nov 2017 – Mar 2023*

- GPA: 3.95/4.0, First Class Honours (Rank 3/100)

Chartered Institute of Management Accountants (CIMA) UK
CIMA Diploma in Management Accounting (CIMA Dip MA) *Nov 2016 – Feb 2018*

PUBLICATIONS

- **Promptception: How Sensitive Are Large Multimodal Models to Prompts?** *EMNLP 2025 (Findings).* [Paper]
  A 61-prompt evaluation framework across 15 categories measuring prompt sensitivity in 10 LMMs on 3 MCQA benchmarks, revealing up to 15% accuracy deviation from phrasing alone.
- **Towards Calibrating Prompt Tuning of Vision-Language Models.** *CVPR 2026.* [Paper]
  Dual-regularization calibration framework (mean-variance margin loss + text moment-matching) for prompt-tuned CLIP, reducing ECE across 11 datasets and 7 prompt-tuning methods without sacrificing accuracy.
- **Real-time Multiple Dyadic Interaction Detection in Surveillance Videos in the Wild.** *IEEE ICIIS 2023.* [Paper]
  Real-time multi-dyadic interaction detection in crowded surveillance video via a skeleton-based X3D-M + attention.
- **Global Horizontal Irradiance Modeling from Sky Images Using ResNet Architectures.** *IEEE ICIIS 2021.* [Paper]
  Hybrid classification-regression framework combining physics-based sky-condition labeling with condition-specific ResNet nowcasters for solar irradiance prediction.

SKILLS & LEADERSHIP

GenAI: LangGraph, LangChain, LlamaIndex, CrewAI, Model Context Protocol (MCP), LLM APIs, Retrieval-Augmented Generation (RAG), structured prompting, LLM-as-Judge evaluation, Langfuse, multi-agent orchestration

Deep Learning & ML: PyTorch, TensorFlow, OpenCV, Hugging Face, CUDA, distributed multi-GPU inference, PEFT (LoRA, QLoRA), SFT, RLHF, vLLM, AutoML (PyCaret), LightGBM, scikit-learn, K-means, pandas, NumPy

Full-Stack: Python, C, Pydantic, Celery, FastAPI, REST, WebSockets, WebRTC; PostgreSQL, pgvector, Qdrant, Redis, SQL, MinIO; React, TypeScript, H3, Streamlit; Git, GitHub Actions, pytest, Playwright, LaTeX; Docker, Azure

AI Coding Tools: Claude Code, Codex, GitHub Copilot, Cursor

Competitive Programming: IEEEXtreme 15.0 (2021) - Global Rank 161/2403, Country Rank 9/300

Founding President - MBZUAI Consulting Club: Established the UAE's first inter-university case competition and preparatory workshops with BCG & Kearney.